Environmental: Sustainable Luxury Fashion

DES 001: Intro to Design
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TA: Latrell

Team Members:
Ally Schiffmaier
Chloe Smith
Savannah Cline

About the Designers!

Ally Schiffmaier
Hi! I am majoring in Computational Cognitive Science through the lens of climate action and design, as a third-year transfer at Davis, with the goal of working in scientific research in SDG 13.

Chloe Smith
Hi! I am a third-year transfer student majoring in Design. I am focusing on and hope to work within the graphic design and UI/UX field.

Savannah Cline
Hello! I'm a second-year transfer student at Davis, and I'm majoring in Design and Communications. I hope to work in the marketing and graphic design field in the future.

Final Showcase: Chaos in the Courtyard

EXPERIENCE:

Our overall experience of installing our demo, and then pitching it to the audience went great! We used several different pitches, and even gave out one of the shirts to an audience member. A lot of audience members had questions about the logo, which was a great talking point to elaborate further into our project, and how it really is sustainable, and helping the environment which was our goal. Although our project wasn't flashy and had tons of people approaching us, we were still able to reach a lot of people who really loved our prototype!

PROCESS BOOK

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Sponge

Research Brief - Ally Schiffmaier

Part II (Research Brief): Climate Change with Focus on Space Trash within Atmospheric Pollution (written by Allison Schiffmaier)

Key findings and issues:
(Definition) Climate change: According to the United Nations on Climate Action, this concept refers to how the temperature and weather experience changes into a state of permanence due to solar weather, volcanic activity, or most often due to humanity's drive towards innovation and synthetic man-made materials (United Nations, 1).

(Statistics): According to the EPA, since 2023, we have experienced a 0.07 F to 2.14 F rise in temperature (in Fahrenheit) globally, with higher and more volatile peaks within the U.S. specifically at 2.39 F (EPA, 1). These findings show that climate change and global warming is a steady rise over time.

(Connection to topic node): According to MIT News, there is a direct correlation between the climate change issue and how various innovative and exploring satellites are able to function within low orbit. This means that greenhouse gases, specifically, are reducing the amount of volume and capacity in the lower atmosphere (Chu, 1-2).
Relevance: One issue is the two sided blade - how might one improve or thicken the atmospheric layer, and what happens as it thins year by year?

(Space Pollution statistics): According to UNOOSA, space has become a limited resource - meaning that for every innovation such as rockets, space stations, satellites, etc., we lose more capacity for it every year. More than this, due to the special reality of the orbit's pull on gravity, it becomes volatile with increased crashes and deadly shards (35,600 larger shards and over 130 million smaller shards) (UNOOSA, 1-2).

Main research projects or products:
Climate monitoring - a tool to see temperature estimates across the globe over time (NOAA, 1) (EPA, 1-2).
The source is a government scientific website, which ensures stability and relevance
This is helpful in viewing hot and cold spots for temperatures, allowing us to see which areas are more insulted and which have less protection.

Real time debris recorder:
Recent research discussed by the European Space Agency shows that there have been recent breakthroughs on how scientists are able to log temperature and pressure density from spacecraft flights using sensors on vehicles. This device is called the Reentry Breakup Recorder, which could potentially be applied into tracking the volatility of space debris and planetary defense (ESA, 1-2).

Scientific Consensus:
A collection of data from 1980 to today that studies temperature anomalies across NASA, NOAA, Hadley Center, and Berkeley Earth.
Lists all relevant labs and scientific societies who focus on or have made significant contributions towards the concerns of climate change.

Key Questions:

- What is the correlation between space trash and climate change, and what actions on the communal and individual level can improve upon this reality?

- If we can make bio-degradable straws and plastics, then how can we identify materials that meet the durability for space flight and be simultaneously sustainable?

- Is there a way to teach materials in low orbit to break down in a less volatile way?

- What are the parameters and requirements for a material to be degradable, sustainable, and be recycled for reuse?

Research Brief - Chloe Smith

Key findings

Chemical contamination in the ocean increases concentrations of nitrogen and phosphorus which "promotes the growth of algal blooms, which can be toxic to wildlife and harmful to humans" (National Geographic)

Micro plastics, which are "less than five millimeters (0.2 inches) in diameter" (National Geographic) have become part of the food chain up to the food humans eat. These microplastics are so prevalent in the food chain that studies "estimates that Americans are eating, inhaling, and drinking at least 74,000 pieces of microplastic a year" (Gross).

The amount of microplastic located in the ocean was averaged out to be about "20,328 particles per square kilometer, with a peak concentration of 580,000 particles per square kilometer in 1997" (Gross)

In the currents between California and Hawaii, about "1.6 million square kilometers (617,763 square miles)" (National Geographic) of both plastic and microplastic floats in the ocean. This patch is referred to as "The Pacific Garbage Patch".

Main research

The Ocean Foundation's Plastics Initiative works to protect both human and environmental health. They intend to achieve this by influencing "sustainable production and consumption of plastics... through holistic policy approach to reduce plastic production" (The Ocean Foundation)

The Ocean Clean up Foundation works to both prevent more plastic and to clean up the existing plastic in the ocean.The Ocean Clean Up created a method that collects floating plastic by using surface nets to navigate the plastic to a vessel that will pick up and collect the plastic.

In order to reduce plastic consumption, marine-derived bioplastics have risen as a more cost effective solution. These bioplastics can be made using seaweed, shrimp shells, langoustine shells, fish skin, fish scales, red algae, and chitosan from seafood waste. Despite the fact that more research is needed to implement in larger scale, this method can reduce the amount of land needed for terrestrial plant based bioplastic as it would "inevitably compete with the production of food crops, thus jeopardizing food security" (ScienceDirect).

Located in pollution hotspots, the organization SeaBin created a floating rubbish bin that removes pollution. The Seabin allows water to flow through while it "intercepts plastic waste, microplastics, fibres, oils, and fuel residues" (SeaBin). These are then collected into a removable catch bag capable of holding up to 20 kilograms.

Key Questions

What else can be done to reduce the amount of microplastic in the ocean?

How can the Pacific Garbage Patch be cleaned up?

How can we increase the use of bioplastic in order to reduce the amount of plastic in the ocean?

Research Brief - Savannah Cline

Key Findings and Issues:
The rapid growth of waste and landfill globally has reached significant levels, to the point of no return, unless we make an effort to change. This type of waste is often referred to as "Global Municipal Solid Waste", and abbreviated as MSW. MSW is projected to rise from 2.1 billion tons in 2023 to 3.8 billion tons by 2050 (UNEP). This represents an 81% increase in waste in the near future, which could still rise or decrease depending on the efforts made.

With the rapid growth of waste comes a rapid increase in microplastics being found in said waste. In France, microplastics are being found in almost every soil sample in their farms (Vincent). The rapid spread of microplastics threatens the biodiversity of food and food systems. Microplastics are also being found in the soil at vineyards, which means the grapes used to make French wine are being contaminated by microplastics as well.

Chemical emissions from landfills in the United States are becoming increasingly more dangerous and are turning into forever chemicals, also referred to as PFAS. These PFAS latch themselves onto groundwater and even our air (Carrington). This can lead to increased long-term health hazards not only for the environment but also for humans.

Research Projects and Products:
One of the most popular projects used to reduce landfill globally is "The Great Global Cleanup," which is featured on Earth Day Network, and has been ongoing since 2023. This is a cleanup campaign that is on a volunteer basis. They organize local groups to remove litter from the streets, parks, and beaches worldwide. It has been reported to collect over 100 million pounds of waste. (Earth Day)

With the rise of AI, there is a ton of "AI-powered" applications, sensors, and garbage sorters, but with the use of Artificial Intelligence comes even more harm to our environment. With the Litterati Global app, people are able to self-report litter in their local areas globally. This app uses AI to generalize the type of litter to getting rid of waste in our communities (Litterati).

Alongside Litterati come similar projects such as "Adopt-A-Street," which is a government initiative where communities can "adopt" an area to clean up. In major cities such as Seattle, Washington, and Austin, Texas, communities have reported significant reductions in waste and litter after this initiative came out (Keep Austin Beautiful).

Key Questions for Future Design:
-How can we optimize motivation or drive for regular, everyday citizens to pick up litter?
-What alternatives are available to stop relying so heavily on landfills for trash?
-How can AI-Powered tools be used to optimize recycling efforts, without further harming the environment?

Mood Board - Ally Schiffmaier

Image Credit: Me @ UCB Stadium
Gloomy weather + orange roadwork sign along greenery to demonstrate emotional theming

Image Credit: Science News Explores
A vivid display of trash in the forest. The article explains this to be small compared to the 100,000-times more trash in the atmosphere

Image Credit: AP News
Rockets are an example of objects or debris entering higher altitude for a study of orbit, but the article goes to show the reliance on the thinning atmosphere to erase the trash

Image Credit: JPL @ NASA
Cube satellites are the recent focus for inexpensive remote space innovation that also a common example of space trash impacted.

Image Credit: Me @ UCB Redwoods
Grounded in nature, shows a deep connection from Earth to Space as well as the refocus of preservation

Image Credit: MIT News
A visual highlight of the impact of space debris, with the source discussing a recent laser breakthrough that allows researchers to see the spread quickly

Mood Board - Chloe Smith

The vessel built by The Ocean Clean Up to remove plastic in the ocean. This helps remove floating plastic from the Ocean

Automated drone, dubbed "Roomba for the Ocean", the WasteShark created by RanMarine which helps remove floating plastic on a smaller scale

The surface nets redirecting floating plastic to a single location, The Ocean Clean Up's vessel

A floating barrier which utilizes the oceans currents to collect floating plastic before it travels farther into the ocean

SeaBins used to collect any floating waste, oil, and microplastics in rivers before entering farther into the ocean

Bioplastics that can be made to reduce the amount of plastic in the ocean as well as decompose faster

Mood Board - Savannah Cline

Trash Formed into the Earth as a play on the Design and double meaning.

Using the Earth as an ashtray serves as a double ontandra.

Informative infographic for how to clean up litter on campus.

A more light-hearted and humorous take on spreading awareness.

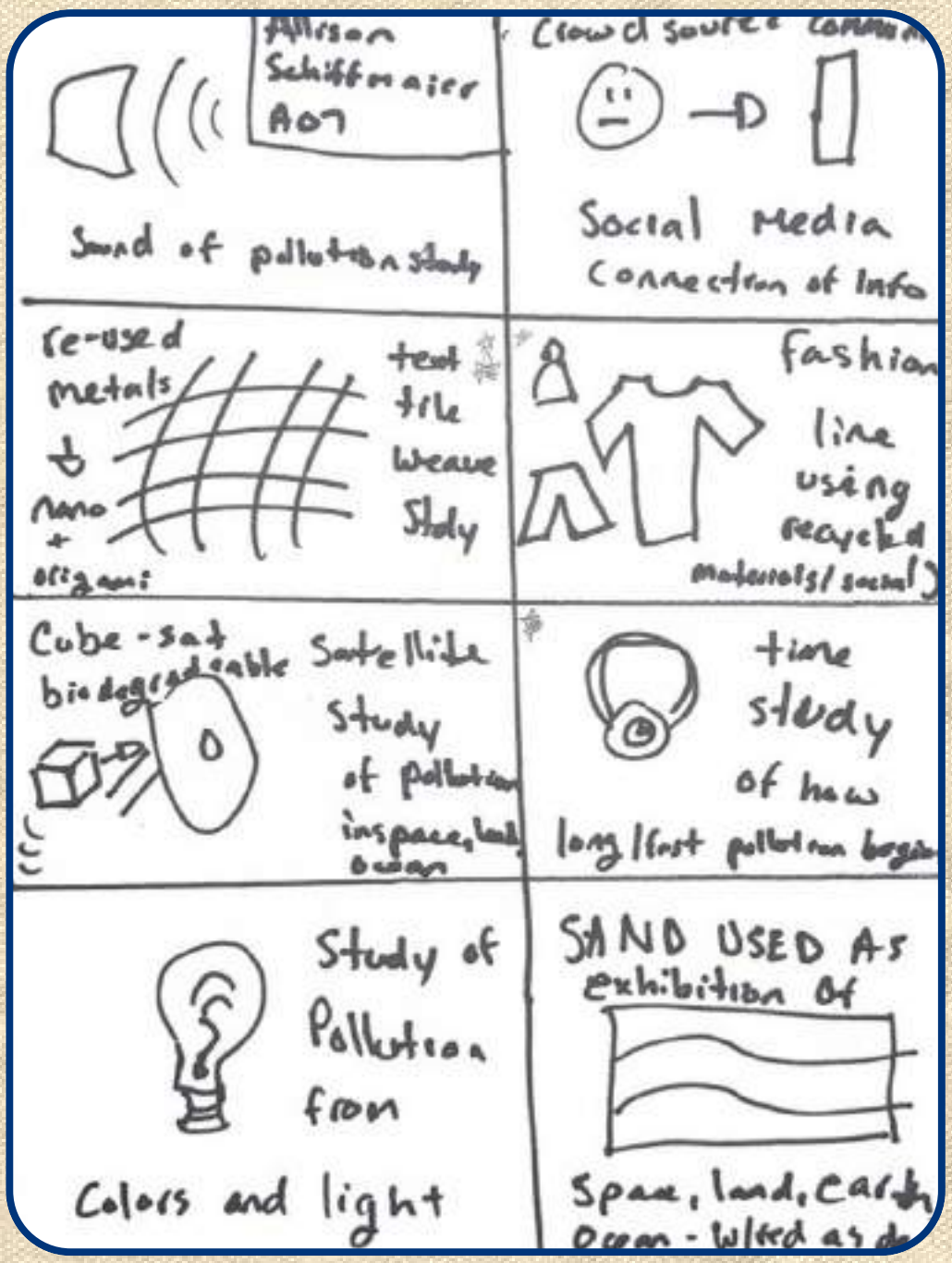
A funny campaign specifically for the beach, catches the eye.

This one is very informative, whilst also being interesting, and has a nice play on words for the caption.

Theme Exploration Map

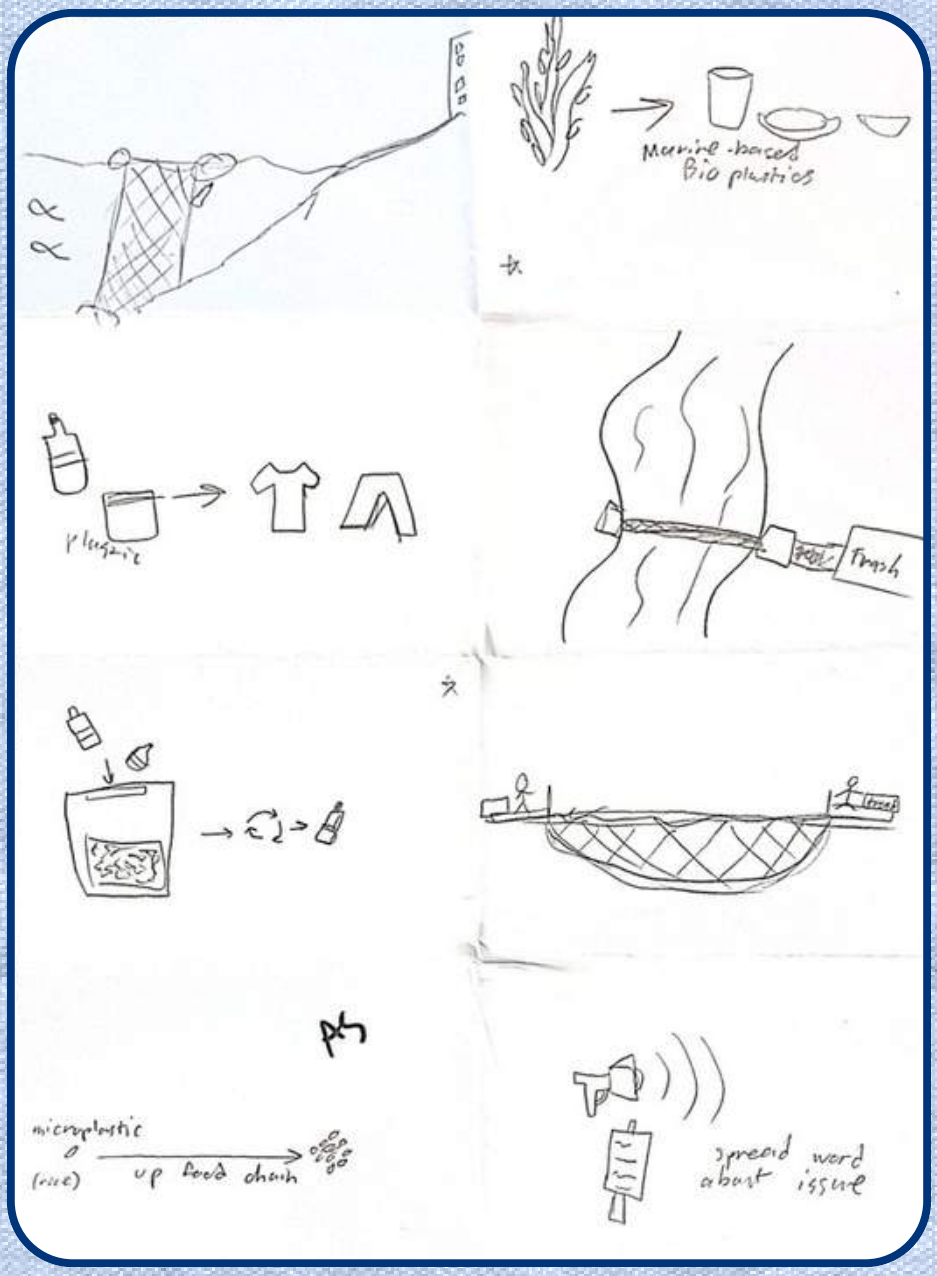
Filter I

Crazy 8's: Ally Schiffmaier



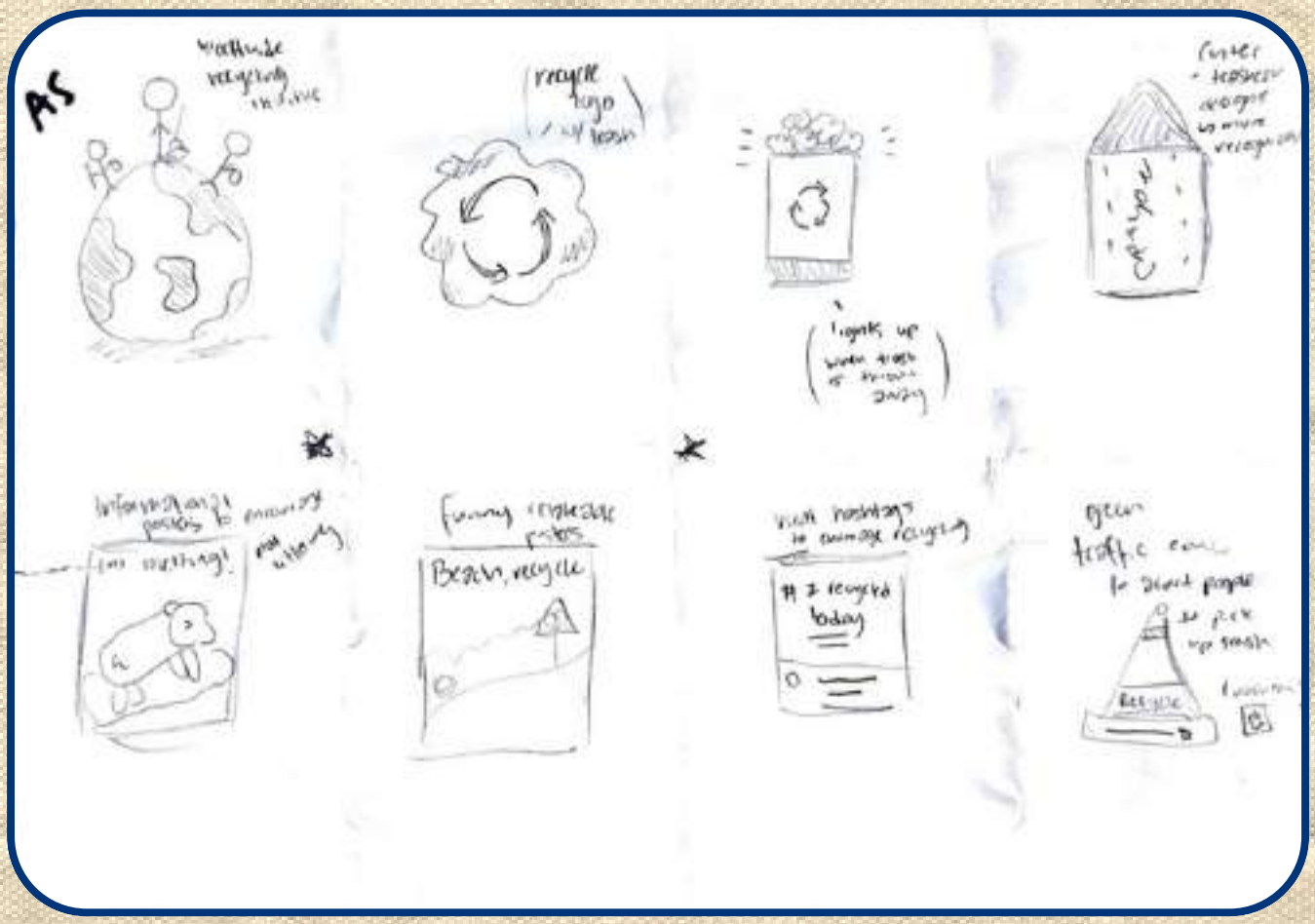
1. A Study of pollution using sound
2. Social Media Campaign to encourage sustainable practices
3. Reuse plastics/old fabrics/materials for second life reuse
4. Make a fashion line of t-shirts, hats, hoodies for sustainable social justice
5. Using cube satellites to study pollution in land, sea, space
6. A live time study of how long it takes a flower or plant to decay near bioplastics verses without bioplastics
7. A study of light pollution through a selection of ultra light color wheel
8. An exhibition of three layers of sand and light poles to illuminate how a fourth strand impacts it

Crazy 8's: Chloe Smith



1) A net to catch trash from entering the Ocean
2) Reusing plastic and turning it into clothing
3) Recycling old plastic bottles into new bottles
4) Demonstrating the amount of microplastic buildup within the food chain using rice
1.5) Turning marine plants into biodegradable objects
2.6) A net placed across a river to catch floating plastic with a place to collect it on the side
3.7) A shallow net that catches plastic to be then collected
4.8) Spreading awareness of the issue of plastic pollution

Crazy 8's: Savannah Cline



1. Sustainable Global Campaign
2. Recycling Inititive Logo
3. Light Up Recycling Can
4. Interactive Trash Cans
5. Informational Poster
6. Funny Relatable Posters
7. Viral Hashtags on Socials
8. Traffic Cones for Recycling

Feedback

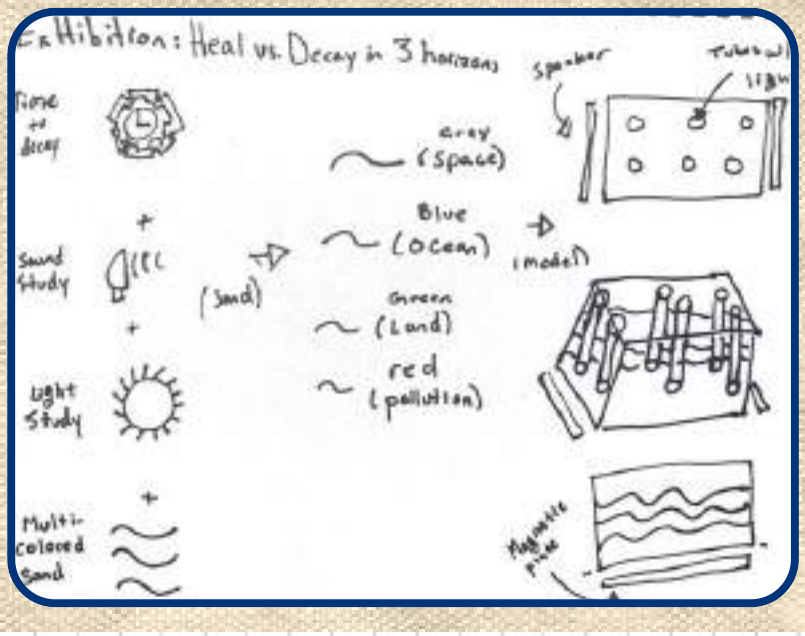
- World wide recycling initiative
- For time/light/sound study
- Utilize rice to mimic pollution
  - Ocean: blue
  - Land: Green
  - Space: Gray
  - Pollution: Red
- Maybe use sand as well.

Sketch and Rational: Ally Schiffmaier

Climate change is impactful on multiple levels - how materials are formed, naturalizing them into something biodegradable for the environment, anticipating budget costs for heating and air during less balanced seasons, wild life, how garbage is processed, susceptibility to wildfires, tsunamis, and earthquakes, and even the ozone layer in the atmosphere. We feel it as the steady climb of temperature during summer with blazing heat in California. This is dangerous on an individual level - but also from the case of the larger community and the environment - largely resulting from irresponsible handling of waste. So what solutions can be constructed to resolve this to the public?

In accordance with the "Climate Change Resources" organization, there have been several works of art and museum exhibitions that demonstrate the impact of climate change on society in finite areas of the world. For example, Musee Magazine's "Asia Society: Coal + Ice." had a focus on the natural icy landscape in Asia taken by a crew of 30 photographers grants a haunting look into the beauty that we should protect. In another example, Ben Miller's "Endangered Rivers" work in Oxbow Gallery shows the beauty of rivers impacted by the environment. (Climate Change Resources, 1) (Musee Magazine 1-2) (Oxbow Gallery 1-2). These two exhibitions of art demonstrate the need to address climate change quickly, given both the public health and emotional aspects as pollution leads into the destruction of life and beauty.

Therefore, our solution is one along the lines of spatial art - exhibition, an intuitive, interactive structure called the "Heal vs. Decay in 3 Horizons" exhibition - a multi-colored sand structure dividing land (green), ocean (blue), and space (grey) with a fourth element called decay (red) - naturally using both a magnetic plate for the interaction element, with harmonic music to show the symbolic comparison of natural beauty when at rest from pollution.

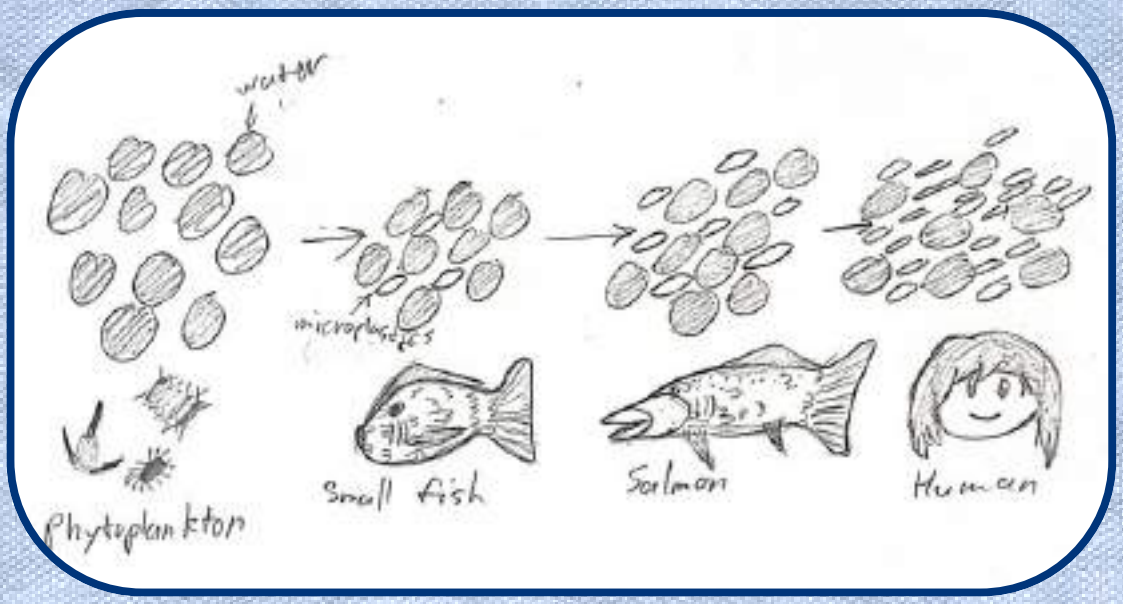


Sketch and Rational: Chloe Smith

With all the plastic in the oceans, it will never truly decompose, simply reducing in size more and more becoming microplastics. As these microplastics enter the food chain, sealife ends up consuming these microplastics. If that was not bad enough, plastics will pick up any toxic pollutants due to the plastics "affinity for chemical sorption" (Ocean River). All these microplastics will eventually end up accumulating in human drinking water and any sea food humans consume. This accumulation of microplastics will lead to yet fully understood long-term damage. So what solutions are there to reduce the amount of plastic from entering the ocean?

According to Columbia Climate School, scientists believe that there are about "24.4 trillion pieces of microplastics... weighing between 82,000 and 578,000 tons". These microplastics deplete the amount of oxygen within seawater as the degradation requires oxygen. This causes "excess nitrogen and phosphorus in seawater... [resulting in] oxygen depletion (Texas Disposal). This depletion will create dead zones as sea life is unable to survive in these low oxygen zones. On top of this, microplastics are consumed by small organisms which will then be consumed by larger animals. As it travels through the food chain, these microplastics will eventually become part of seafood that humans eat. As plastic can absorb toxins, an example is a toxin called methylmercury "(bacteria-converted mercury carried down to the ocean's surface)" (Texas Disposal) being absorbed by phytoplankton. This toxin will travel up the food chain where once reaching fish that is about the size of swordfish, the mercury concentration will be very high.

In order to reduce the amount of plastics entering the ocean, we must bring awareness to the amount of microplastics that are currently present in the ocean. To do this, our solution is to demonstrate how microplastics accumulate over time. By showing how the amount of microplastics build up as it travels up through the food chain, entice others to begin recycling as to reduce the amount of plastic that ends up in the ocean.

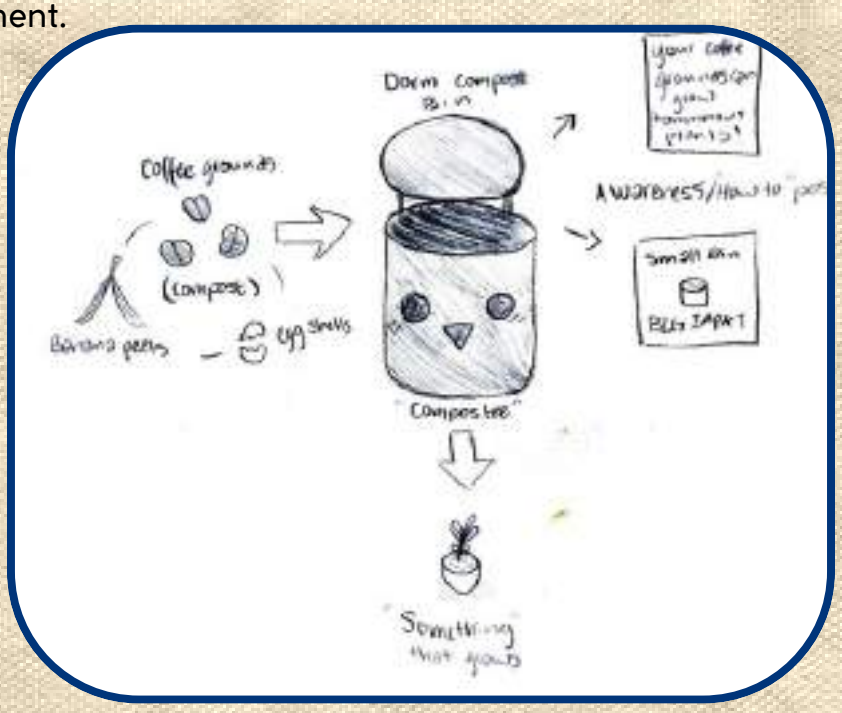


Sketch and Rational: Savannah Cline

Action must be taken, whether it be small or big; we have to make a change for our environment. As college students, there is only so much we can do with our low budgets, busy schedules, and overall exhaustion from school and work. With the Compostee comes a simple solution to a much bigger problem. Living in dorms, or tight spaces with roommates, makes it extremely difficult to compost old fruits, coffee grounds, eggshells, and basically any old food. The results of not taking a few seconds out of our day to compost come with bigger problems, such as greenhouse gas emissions, which are actively harming our environment.

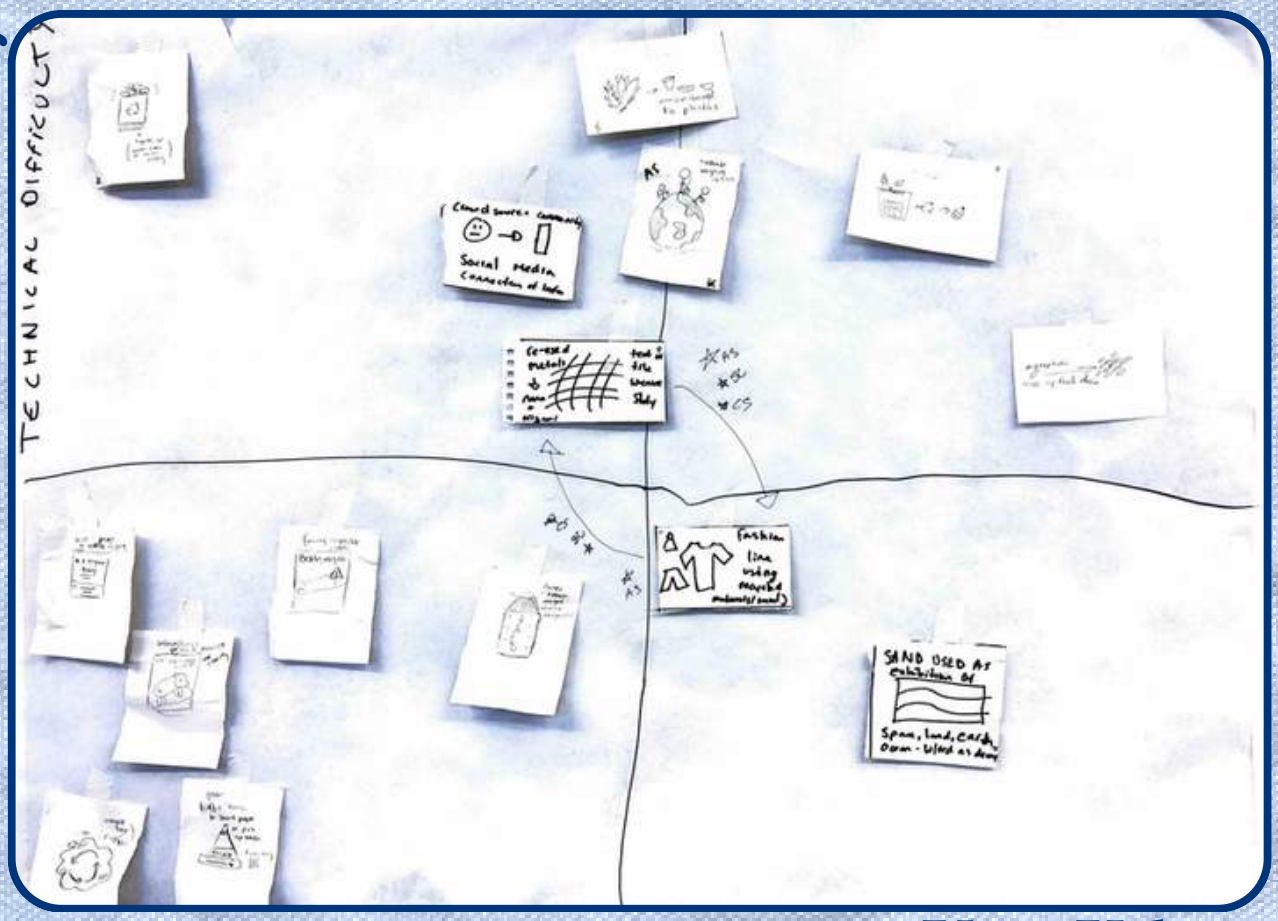
Composting is an extremely important part of contributing to protecting the environment. According to the EPA (Environmental Protection Agency), Composting can significantly reduce the amount of trash that is in landfills, and create healthy soil to further help our environment. UC Davis is already taking several initiatives to compost and create community gardens to help the environment, but it can be taken even further. According to the UC Davis Student Housing and Dining Services, the "Project Compost" has been around since 1999 as a recycling initiative, and collects compost from several different locations on campus, and then takes it to the student farm to be used in gardening and planting efforts. Another similar project happens at our neighboring school, Sacramento State, and their environmental inititive BAC Yard. BAC Yard is able to compost up to 100 cubic yards of material each year according to their website.

With initiatives like these, they prove that ideas like Compostee can work when applied in the right areas. I believe that with Compostee, UC Davis, and campuses around the world would be able to reduce green house gas emissions significantly, and overall help save the environment.



Decision Matrix

Technical Difficulty



User Value

During the Decision Matrix, we organized all of our favorite design ideas. We chose Sustainable Fashion as a middle-ground between user value and technical difficulty.

We chose to lean into sustainable fashion - but make it luxurious. We decided we wanted to use sustainable materials, without sacrafising the quality.

Moodboard

Sustainable Luxury Clothing

Moodboard Analysis

Overview:

Our mood board explores how both luxury and sustainability can coexist alongside high-end fashion. We used visual elements to emphasize luxury, royalty, and overall a high-end feeling to the clothes and overall theme. Each image represents a different aspect of the sustainable luxury idea, ideals of eco-friendly production of clothes, royal luxurious aesthetic, and overall excellence in craftsmanship for sustainable fashion.

Analysis:

Making Consumers Aware (Recyclable / Sustainable Materials used tag)
The image overall represents how to showcase items that are made with sustainable materials, and to communicate that to consumers. This inspires us to add to our fashion line that our clothes are made of sustainable materials. Our design will differ by using a more luxurious design and sustainable materials that are biodegradable.

Modern Luxurious Sustainable Fashion
The image showcases a model wearing a gender neutral luxurious blazer. This inspires us to combine a gender neutral look with sustainable features. Our design will differ by providing a more sustainable option and longer longevity.

Conveying a Sustainable Message
The image represents a message and movement promoting both sustainable fashion and spreading awareness for sustainability. This inspires us to incorporate more features to better promote sustainability. Our design will differ by using more luxurious design methods and elements.

Tailored Sustainable Design
The image is a representation of tailoring our fashion to our consumers. This inspired us to make sure consumers have a wide variety of sizing and tailoring options. Our design will differ by offering more advanced sustainable materials, such as organic cotton blends, to ensure the garments last longer.

Everlane Sustainable Fashion
This image represents our main inspiration source - Everlane, which is a sustainable fashion brand. This inspires us to look into more sustainable factors, such as using organic cotton, recycled polyesters, and recycled leather. Our design differs by combining sustainable materials with luxurious design.

Recyclable Materials
This represents a symbol for encouraging recycling and is a visual communication tool. This is inspiring to us because it helps us keep in mind how to visually communicate sustainability effectively by using universal symbols. Our design will differ by using a more luxurious tone for universal symbol use within our design.

Fashion Inspiration
This depicts overall luxury and fashion elegance. This inspires us to use aspects from common luxurious designs to convey our theme and overall message. Our design will differ by using sustainable upcycled materials rather than fast-fashion methods.

Kanban Board

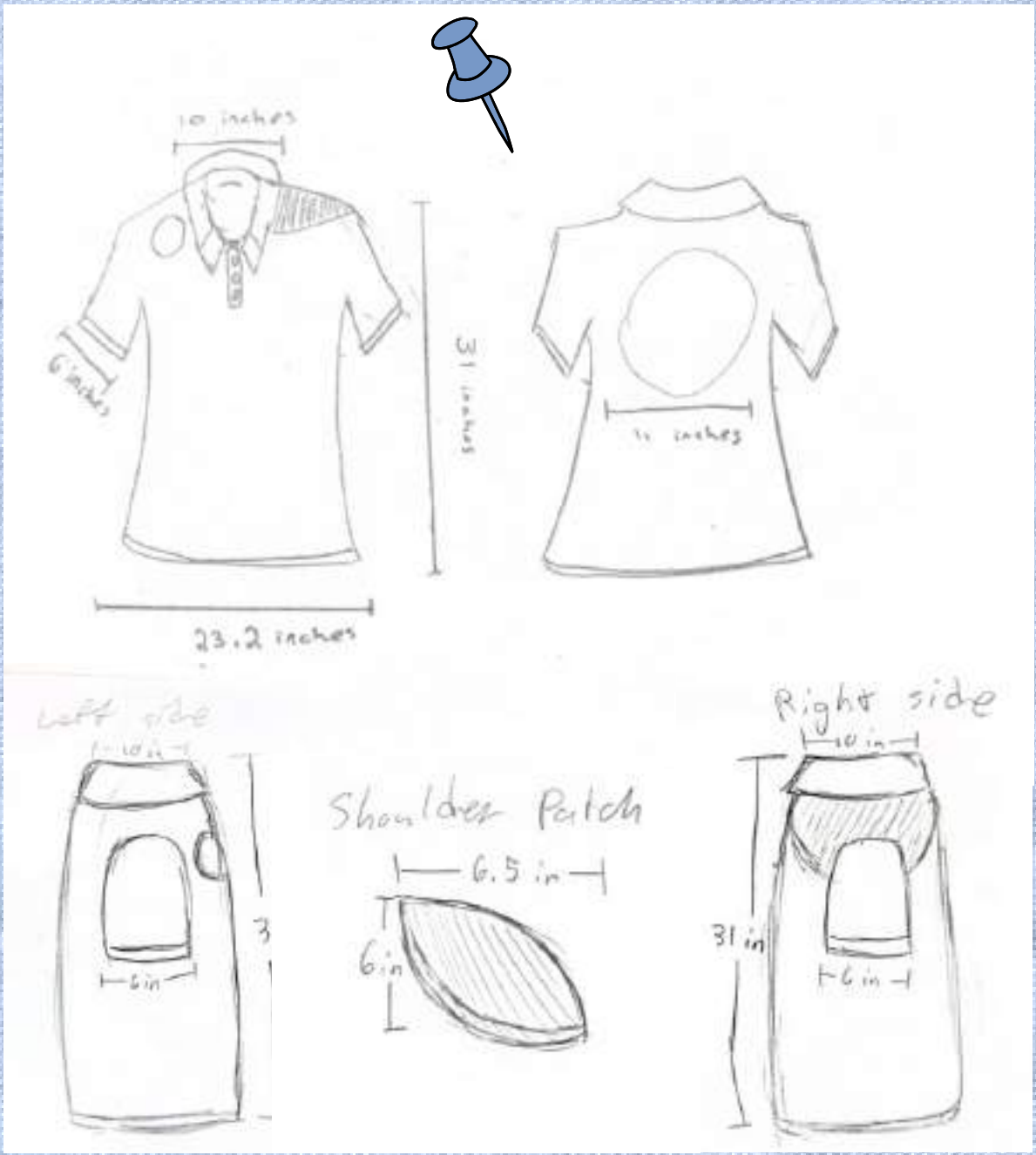
To-Do	In Progress	Completed
Ask DES Faculty about the Print Screen Lab	Full Measurement Sketch	Decision Matrix
Develop photo section for process book to show the lifespan of the project	Determine vendors for fashion line	Kanban Board
Develop ideation purpose for the fashion line for our "why"	Sketch patch design	Bill Of Materials
Research durable materials		Mood Board + Analysis
		Determine Color Palette
		Develop a consistent theme for the Process Book
		User perspective sketch
		Group Prototype

Bill of Materials

Material/ Equipment	Quantity	Unit	Link/Catalog/Source	Cost	Estimated Arrival Time	Team Member
Shirt Printer	12		Ethan Wade Graphics	~$350	1.5 Weeks	Ally Schiffmaier
Polo Shirt	1	Size XL	Walmart	$18	1 Day	Ally Schiffmaier
Fabric	1	2 yards	Walmart	$6	1 Day	Ally Schiffmaier
Sewing Thread (Bundle)	1	25 yd each (12 spools)	Target	$6	1 Day	Chloe Smith
Sewing Kit	1	62 Pieces	Target	$11	1 Day	Chloe Smith
Old Navy Shirt	2	Size XL	Old Navy	$23	1 Day	Ally Schiffmaier

Voice I

Product Dimensions Diagram



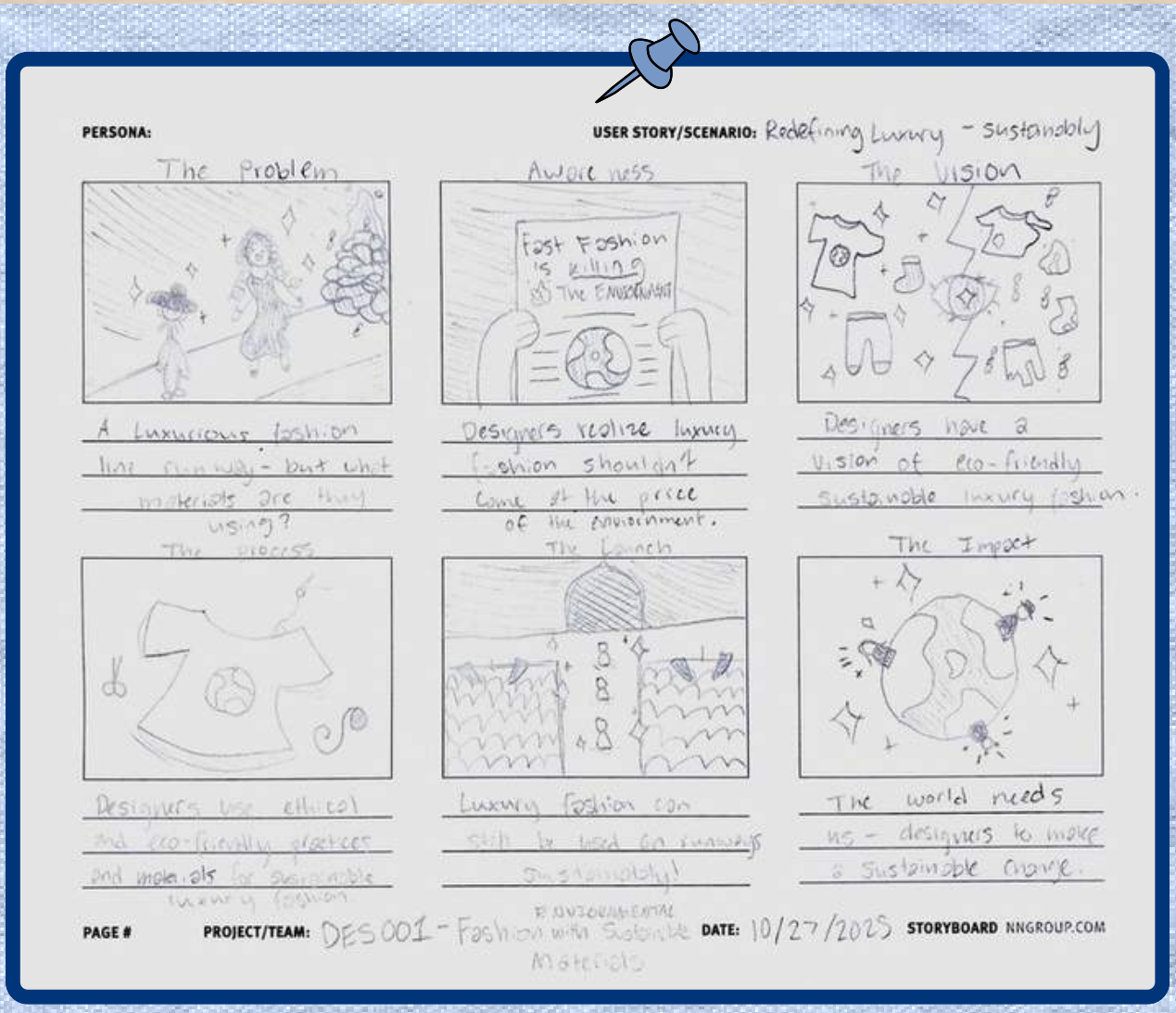
Dimensions:

- Length: 31 inches
- Width: 23.2 inches
- Sleeve: 7 inches around (est)
- Collar: 10 inches around (est)

User Interaction: Ally Schiffmaier

A scenario of a user analyzing our polo design and buying it in a UC Davis store.

Storyboard - Savannah Cline

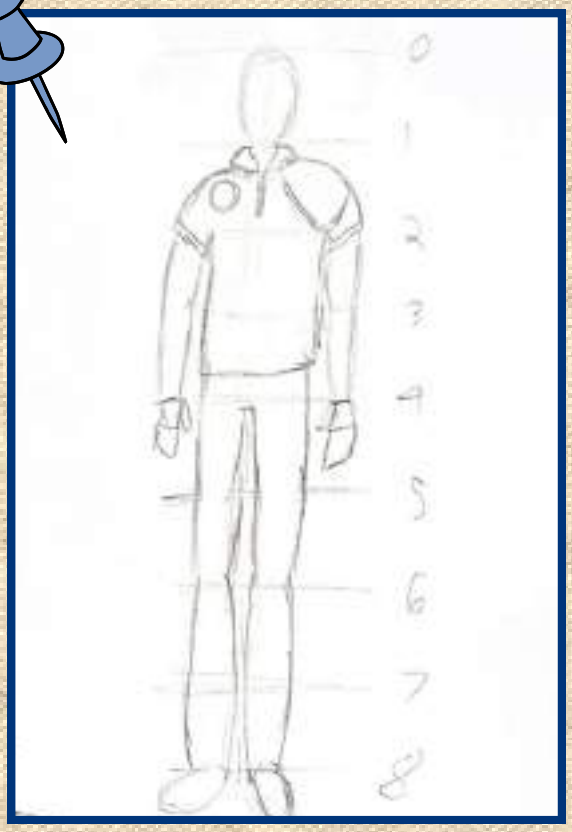


The storyboard reflects on how designers have realized that while luxurious fashion is beautiful, it is coming at the cost of killing the environment.

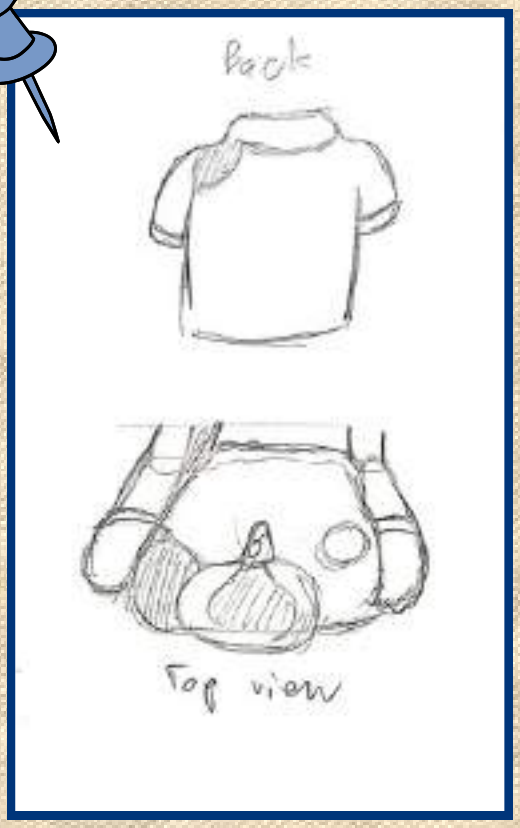
Designers come up with the idea of sustainable luxurious fashion - and use sustainable, and ethical materials and work practices to produce this.

Luxurious fashion can still be beautiful, while keeping our planet beautiful simultaneously.

Perspective View: Chloe Smith



What the shirt will look like on a person.



A back view of the design as well as a first person top down view.

What the design looks like at different angles.

Voice II

Prototypes

Prototype 1

This is the second/remade version of the polo prototype using a hand-sized cardboard tube, a table napkin for the shirt material, a small ribbon to tie it together, and a mechanical pencil for the arms. I made small incisions for the arm cut-outs so the pencil could splice through for the arms. Then, to fit around the polo material, I measured where it hit on the material in the fit, and cut there. Once this was finished, I wrapped it around the tube and tied it with the ribbon. Finally, I cut and folded the top for the polo collar, and made dots on the layer beneath the collar to simulate the shoulder patch.

Prototype 2

The core feature is the logo that will be present on the shirts. The location of the logo will be located on the right side, a few inches from the shoulder. The design has the direction of each element trying to follow the same direction as the recycling logos. I incorporated features that symbolize the land, ocean, and space with the tree, waves, and stars, respectively, as a reference to the research done in the beginning. To refine it more in the future, instead of colors, the logo will have more simplistic outlines of each of the features, rather than the multiple colors present.

Prototype 3

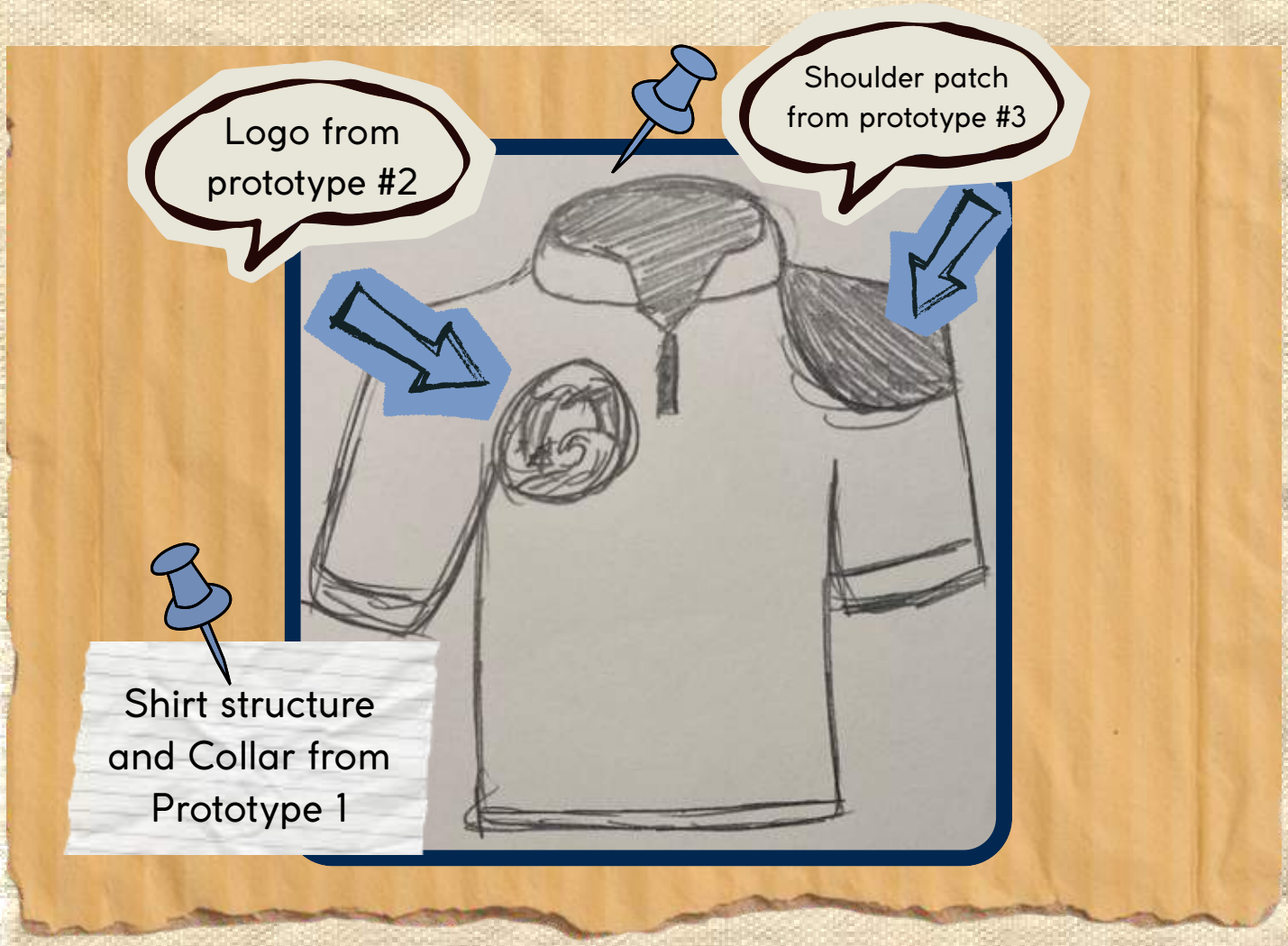
This shoulder patch design is inspired by knight's shoulder armor, but modernized and made sleek for the polo design. However, instead of armoring for battle, this patch will "armor" for weather conditions and elevating the fashion in the room. This prototype version is made with elegant tissue paper for it's malleability, with the lines made to notate the seams where the manufacturer will attach it to our polo design, alongside our logo design in prototype #2. This shoulder patch will be made of suede or velvet texture to deepen the luxurious overall feel of our polo.

Prototype Decision

After reviewing all of the prototypes we have created, the final design will combine the best and most effective elements from all of the prototypes. We went through a lot of trial and error, and had to do a lot of prototyping to come to our conclusion in our prototype process. For prototype 1, it was successful in demonstrating the overall polo's basic structure, and the collar formation based on the shoulder structure/design. Alongside prototype 3, we can see a more blown-up version of the shoulder patch that would be on prototype 1. We used this hybrid to overall capture the everyday comfort of a polo, while being elegant and luxurious sustainable fashion.

For future improvements, the polo will include the logo from prototype 2. This logo overall tells the story behind our design purpose, and adds a pop of color without sacrificing the overall integrity of the shirt. For future improvements, we will experiment with texture, material, and color for all of these prototypes, while maintaining the same shapes and overall structure as provided in the prototypes.

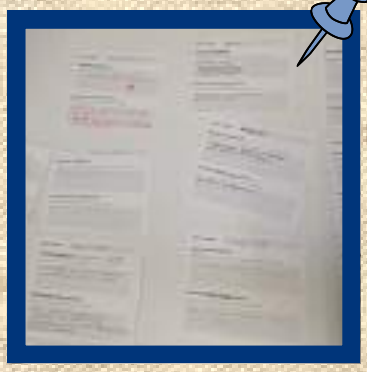
Annotated Sketch

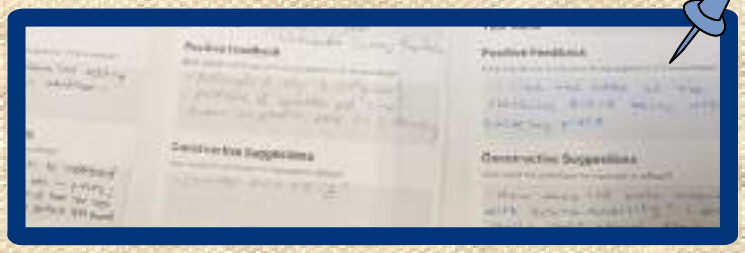


The final prototype in the annotated sketch combines all elements; the shirt structure, and collar from prototype 1, the logo concept from prototype 2, and the shoulder patch from prototype 3. In this polo, we chose each element carefully, to create a design that balances function, and fashion. Each of these elements was chosen to create our final, polished sketch, and will be in our final version of the shirt. The collar adds a professional touch, while the shoulder patch adds flair and character, and the logo overall tells our story, and pulls the design together visually, and symbolically. It's all tied together with the logo to reflect our message; about being sustainable, and how to save the enviornment one shirt at a time. Overall, it achieves the design goal of an elegant, yet sustainable garmet.

Voice III

User Study Report





UI Improvement Feedback:

- Add diagram to final table for parts
- Make an info-card to explain inspiration behind the design
- Clarify on how it is sustainable

UI Appreciation Feedback:

- Loved that it is a conversation piece
- Symbolism in social climate justice is powerful
- The shoulder patch adds depth and luxury

Actionable Tasks

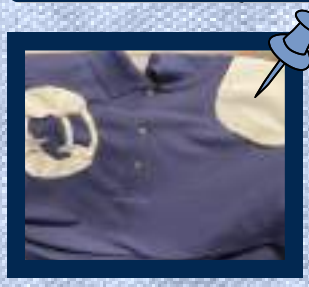
- Buy polo — 1% (Ally) - Done
- Buy microsuede — 1% (Ally) - Done
- Buy thread — 1% (Chloe) - Done
- Stitch fabric / initial sewing work — 50% (Chloe) - Done
- Take process photos — 2% (Ally) - Done
- Create cards/tags for polo (environment, carbon footprint card, assembly card, open-source "how-to" shirt guide) — 25% (Savannah) - Done

- (All completed by 11/19/25)

- Compile final design components (graphics, materials, sizing) — 2% (All) - In Progress
- Determine vendors for final manufacturing — 2% (Ally) (Not Started)
- Order final render — 5% (Ally) (Not Started)
- Pick up order & conduct final photoshoot — 5% (Ally, Chloe) (Not Started)
- Procure model for highest fidelity prototype — 2% (Ally, Chloe) (In Progress)
- Package final presentation materials to Chaos Courtyard — 2% (All) (Not Started)

- (All due by 12/2)

Implementation Process



Development on Polo

- Materials: Ally obtained and bought the sample polo and patch material (velvet was too expensive, then swapped it with micro-suede), then delivered to Chloe to construct
- Construction: Chloe cut and sewed the hero design (water, tree, star) and the shoulder patch onto the polo
- Modeling: Ally procured a model (Citali) for the polo at TRC, and took the photo
- Highest Fidelity Research: Ally is actively looking into local resources for a polished prototype. The main methods are embroidery with patches, screen-print, and iron-on for each element.

Implementation Process

ENVIORNMENTAL FACTS CARD

- Organic & regenerative cotton: grown without synthetic pesticides or fertilizers
- Recycled polyester & nylon: reduces waste and energy consumption
- Plant-based dyes: minimizes toxic chemical runoff
- Carbon-neutral production: greenhouse gases offset through verified programs
- Zero-waste pattern cutting: maximizes fabric efficiency and minimizes scraps
- Cards: Developed by Savannah

Voice IV

Design Style: Ally Schiffmaier

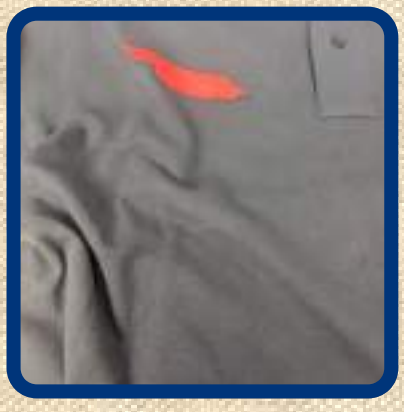
Materials

70% fidelity Polo Product

- Materials: 100% Polyester
- Walmart
- Sewn in patches with cut micro suede for hero image + shoulder patch

85% fidelity Polo Product

- Material %: 20% cotton, 80% polyester
- Amazon for printing + polo
- Sent design after identifying seller



100% fidelity Polo Product

- Material %: 40% cotton, 60% polyester
- Old Navy, Ethan Wade Graphics
- Completed final hero design(s), then shipped to Ethan Wade for rapid iron-on printing

Design Style: Chloe Smith

Icons

Front Logo:
Created with the intention to address pollution that is located in the oceans, on land, and in space and was made with the recycle symbol in mind.

Back Logo:
Created to identify our product and our goal on sparking conversations on climate change and action using what we wear

Design Style: Savannah Cline

Color Palette

(Primary) **Royal Blue - #00357a**

- Used for the primary & base color of the clothing & overall branding.

(Secondary) **Khaki - #f1ce8d**

- Used for the tag & Icon, as well as the overall secondary color of the branding.

(Accent) **Pure White - #FFFFFF**

- Used for the accent pieces of the clothing, as well as accent colors in the branding.

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Appendix

Assignment:	Coordinator:	Compilation Date:
Sponge II	Chloe Smith	10/15/25
Filter I	Savannah Cline	10/21/25
Voice I	Ally Schiffmaier	10/29/25
Voice II	Savannah Cline	11/5/25
Voice III	Ally Schiffmaier	11/19/25
Voice IV	Chloe Smith	12/1/25